

KNYSNA, South Africa

WMO: 689350

Lat: 34.0490S

Lon: 23.0814E

Elev: 177

StdP: 14.60

Time Zone: 2.00 (E02)

Period: 98-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	42.8	44.6	34.6	29.5	60.5	37.1	32.7	55.6	19.8	55.5	17.1	57.3	0.5	170	0.297

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	13.0	86.7	68.2	83.1	68.1	80.3	67.6	72.6	80.9	71.3	78.7	70.2	77.0	5.6	230

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
70.1	111.3	76.6	68.9	106.7	75.3	67.8	102.8	74.2	36.4	80.9	35.3	79.2	34.3	77.0	80.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 16.0	13.0	10.9	40.2	97.4	1.4	3.1	39.2	99.6	38.5	101.4	37.7	103.1	36.7	105.4		
(5)			38.7	75.4	1.3	2.4	37.8	77.1	37.0	78.6	36.3	79.9	35.3	81.7		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	64.0	71.1	71.7	69.2	65.2	61.7	58.0	57.2	57.5	59.7	63.1	65.5	69.1
	DBStd	6.62	3.66	3.81	4.20	4.71	4.38	4.44	4.73	4.20	4.03	4.40	3.84	3.78
	HDD50	2	0	0	0	0	0	0	1	1	0	0	0	0
	HDD65	1191	2	1	9	52	122	218	249	238	167	88	37	8
	CDD50	5128	653	608	596	456	362	239	225	233	290	407	466	593
	CDD65	843	191	189	139	58	19	7	8	6	7	30	53	136
	CDH74	3293	710	705	482	251	140	67	76	40	56	151	147	468
	CDH80	736	137	162	112	65	36	17	15	9	19	49	26	89
Wind	WSAvg	3.5	3.8	3.5	3.3	3.0	2.9	3.1	3.3	3.6	3.9	4.2	4.0	4.0
Precipitation	PrecAvg	28.5	2.0	1.8	2.4	2.6	2.4	2.4	2.1	3.0	2.4	3.0	2.4	2.1
	PrecMax	45.4	5.9	4.9	8.3	8.8	7.4	7.4	6.0	8.7	9.5	8.2	6.3	6.6
	PrecMin	21.0	0.6	0.4	0.2	0.5	0.2	0.3	0.2	0.4	0.2	0.4	0.3	0.0
	PrecStd	5.5	1.1	1.0	1.5	1.8	1.6	1.8	1.4	2.0	2.2	2.0	1.4	1.4
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	88.5	91.3	89.4	87.0	84.5	82.7	82.1	79.1	81.4	87.8	83.6	87.4
		MCWB	71.1	73.3	70.2	66.9	62.1	60.1	59.3	58.7	61.6	66.0	66.6	68.4
	2%	DB	83.4	84.3	82.8	80.4	77.5	74.7	74.9	71.5	73.5	77.9	78.0	81.5
		MCWB	70.4	70.8	69.0	66.1	62.2	58.1	57.4	58.5	60.9	63.8	66.6	69.0
	5%	DB	80.1	80.4	79.0	75.9	72.2	69.8	69.0	67.6	69.4	73.1	74.6	78.5
		MCWB	69.4	69.4	68.4	64.8	61.4	57.9	56.7	57.3	59.4	63.0	65.4	67.9
	10%	DB	77.7	77.8	75.9	72.5	68.9	66.2	65.3	64.7	66.4	70.0	72.0	76.1
		MCWB	68.4	68.7	67.2	64.1	60.4	57.1	56.1	56.3	58.3	62.0	64.2	67.1
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	74.0	74.9	72.8	70.1	67.4	63.1	62.4	63.0	64.8	68.7	70.7	73.2
		MCDB	82.9	85.6	81.9	80.2	77.0	73.3	72.7	72.1	74.3	79.8	77.4	80.9
	2%	WB	72.2	72.7	70.9	68.2	64.5	61.0	60.0	60.6	62.5	65.8	68.5	71.2
		MCDB	80.2	81.0	79.0	76.7	72.6	69.4	68.7	68.0	69.3	74.4	74.6	78.0
	5%	WB	70.8	71.3	69.6	66.6	62.7	59.3	58.3	58.9	60.9	64.2	66.8	69.7
		MCDB	78.1	77.9	76.9	73.6	69.7	67.7	66.3	65.6	67.6	71.2	72.7	75.9
	10%	WB	69.6	70.0	68.4	65.0	61.1	57.8	56.8	57.3	59.2	62.8	65.3	68.2
		MCDB	76.2	75.9	74.6	71.0	67.4	65.2	64.1	63.7	65.5	68.7	71.0	74.9
Mean Daily Temperature Range	5% DB	MDBR	12.8	13.0	13.3	14.5	15.8	17.8	16.9	15.6	13.8	12.9	12.4	13.0
		MCDBR	17.4	17.8	19.1	21.2	22.1	24.2	23.7	21.4	19.8	18.9	17.2	17.2
		MCWBR	7.7	7.5	8.2	9.6	11.0	12.4	11.9	10.9	10.2	9.1	8.4	7.9
	5% WB	MCDBR	14.8	14.9	15.7	17.9	18.4	20.3	19.6	18.7	17.8	16.6	14.6	14.3
		MCWBR	7.1	7.0	7.5	9.0	10.6	12.0	11.5	10.8	10.2	8.8	7.8	7.3
Clear-Sky Solar Irradiance	taub	0.381	0.385	0.371	0.344	0.329	0.317	0.319	0.350	0.366	0.387	0.371	0.378	
	taud	2.477	2.471	2.497	2.556	2.574	2.582	2.570	2.473	2.410	2.386	2.468	2.472	
	Ebn at Noon	303	295	288	279	266	261	266	272	284	291	305	305	
	Edh at Noon	37	36	33	28	24	22	23	29	35	39	37	37	
All-Sky Solar Radiation	RadAvg	2364	2114	1727	1293	971	833	910	1173	1585	1930	2271	2450	
	RadStd	89	73	90	56	50	46	56	55	74	100	113	137	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (6 neighbors)	N/A	N/A	+3.38	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page